

```
/* =====  
ECOMMERCE STORED FUNCTIONS LAB  
Microsoft SQL Server Version  
===== */
```

```
IF DB_ID('ecommerce_db') IS NULL  
    CREATE DATABASE ecommerce_db;  
GO
```

```
USE ecommerce_db;  
GO
```

```
/* =====  
DROP OLD TABLES  
===== */
```

```
DROP TABLE IF EXISTS order_items;  
DROP TABLE IF EXISTS orders;  
DROP TABLE IF EXISTS products;  
DROP TABLE IF EXISTS customers;  
GO
```

```
/* =====  
CREATE TABLES  
===== */
```

```
CREATE TABLE customers (  
    customer_id INT IDENTITY(1,1) PRIMARY KEY,  
    customer_name VARCHAR(100),  
    email VARCHAR(100),  
    membership_level VARCHAR(30)  
);
```

```
CREATE TABLE products (  
    product_id INT IDENTITY(1,1) PRIMARY KEY,  
    product_name VARCHAR(100),  
    category VARCHAR(50),  
    unit_price DECIMAL(10,2),
```

```

    stock_qty INT
);

CREATE TABLE orders (
    order_id INT IDENTITY(1,1) PRIMARY KEY,
    customer_id INT,
    order_date DATE,
    order_status VARCHAR(30),
    CONSTRAINT FK_orders_customers
        FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);

CREATE TABLE order_items (
    order_item_id INT IDENTITY(1,1) PRIMARY KEY,
    order_id INT,
    product_id INT,
    quantity INT,
    unit_price DECIMAL(10,2),
    CONSTRAINT FK_order_items_orders
        FOREIGN KEY (order_id) REFERENCES orders(order_id),
    CONSTRAINT FK_order_items_products
        FOREIGN KEY (product_id) REFERENCES products(product_id)
);
GO

/* =====
   POPULATE SAMPLE DATA
   ===== */

INSERT INTO customers (customer_name, email, membership_level)
VALUES
('Aung Aung', 'aung@example.com', 'Gold'),
('Su Su', 'su@example.com', 'Silver'),
('Kyaw Kyaw', 'kyaw@example.com', 'Normal'),
('Mya Mya', 'mya@example.com', 'Gold'),
('Ko Ko', 'koko@example.com', 'Silver'),
('Hla Hla', 'hla@example.com', 'Normal'),
('Zaw Zaw', 'zaw@example.com', 'Gold'),

```

```
('Ei Ei', 'eiei@example.com', 'Silver'),  
( 'Tun Tun', 'tun@example.com', 'Normal'),  
( 'May May', 'may@example.com', 'Gold');
```

```
INSERT INTO products (product_name, category, unit_price, stock_qty)  
VALUES  
( 'Laptop', 'Electronics', 1200000, 10),  
( 'Mouse', 'Electronics', 25000, 100),  
( 'Keyboard', 'Electronics', 45000, 50),  
( 'Office Chair', 'Furniture', 180000, 20),  
( 'Desk', 'Furniture', 250000, 15),  
( 'Monitor', 'Electronics', 350000, 25),  
( 'Printer', 'Electronics', 450000, 8),  
( 'USB Keyboard', 'Electronics', 30000, 120),  
( 'Gaming Mouse', 'Electronics', 55000, 75),  
( 'Bookshelf', 'Furniture', 200000, 12),  
( 'Meeting Table', 'Furniture', 650000, 5),  
( 'Headphone', 'Electronics', 85000, 40),  
( 'Webcam', 'Electronics', 95000, 18),  
( 'Air Conditioner', 'Appliances', 1800000, 3),  
( 'Projector', 'Electronics', 950000, 6);
```

```
INSERT INTO orders (customer_id, order_date, order_status)  
VALUES  
(1, '2026-05-01', 'Completed'),  
(2, '2026-05-02', 'Completed'),  
(3, '2026-05-03', 'Pending'),  
(4, '2026-05-04', 'Completed'),  
(5, '2026-05-05', 'Pending'),  
(6, '2026-05-06', 'Cancelled'),  
(7, '2026-05-07', 'Completed'),  
(8, '2026-05-08', 'Completed'),  
(9, '2026-05-09', 'Pending'),  
(10, '2026-05-10', 'Completed');
```

```
INSERT INTO order_items (order_id, product_id, quantity, unit_price)  
VALUES  
(1, 1, 1, 1200000),
```

```
(1, 2, 2, 25000),
(2, 4, 1, 180000),
(2, 3, 1, 45000),
(3, 5, 1, 250000),
(4, 6, 1, 350000),
(4, 7, 2, 450000),
(5, 8, 1, 30000),
(5, 9, 2, 55000),
(6, 10, 1, 200000),
(7, 11, 1, 650000),
(7, 12, 3, 85000),
(8, 13, 2, 95000),
(9, 14, 1, 1800000),
(10, 15, 1, 950000);
GO
GO
```

```
/* =====
DROP OLD FUNCTIONS
===== */
```

```
DROP FUNCTION IF EXISTS fn_order_subtotal;
DROP FUNCTION IF EXISTS fn_tax_amount;
DROP FUNCTION IF EXISTS fn_membership_discount;
DROP FUNCTION IF EXISTS fn_order_final_total;
DROP FUNCTION IF EXISTS fn_stock_status;
DROP FUNCTION IF EXISTS fn_customer_order_count;
DROP FUNCTION IF EXISTS fn_membership_label;
DROP FUNCTION IF EXISTS fn_shipping_fee;
DROP FUNCTION IF EXISTS fn_order_grand_total;
GO
```

```
/* =====  
FUNCTION 1: Calculate Order Subtotal  
===== */
```

```
CREATE FUNCTION fn_order_subtotal  
(  
    @p_order_id INT  
)  
RETURNS DECIMAL(10,2)  
AS  
BEGIN  
    DECLARE @v_subtotal DECIMAL(10,2);  
  
    SELECT @v_subtotal = SUM(quantity * unit_price)  
    FROM order_items  
    WHERE order_id = @p_order_id;  
  
    RETURN ISNULL(@v_subtotal, 0);  
END;  
GO
```

```
174 SELECT dbo.fn_order_subtotal(1) AS order_subtotal;  
175 GO  
176
```

100 % No issues found

Results		Messages	
	order_subtotal		
1	1250000.00		

```
/* =====  
FUNCTION 2: Calculate Tax Amount  
===== */
```

```
CREATE FUNCTION fn_tax_amount  
(  
    @p_amount DECIMAL(10,2)  
)  
RETURNS DECIMAL(10,2)  
AS  
BEGIN  
    RETURN @p_amount * 0.05;  
END;  
GO
```

```
192  SELECT dbo.fn_tax_amount(100000) AS tax_amount;  
193  GO  
194
```

100 % No issues found

Results Messages

	tax_amount
1	5000.00

```
/* =====  
FUNCTION 3: Membership Discount  
===== */
```

```
CREATE FUNCTION fn_membership_discount  
(  
    @p_amount DECIMAL(10,2),  
    @p_membership_level VARCHAR(30)  
)  
RETURNS DECIMAL(10,2)  
AS  
BEGIN  
    DECLARE @v_discount DECIMAL(10,2);  
  
    IF @p_membership_level = 'Gold'  
        SET @v_discount = @p_amount * 0.10;  
    ELSE IF @p_membership_level = 'Silver'  
        SET @v_discount = @p_amount * 0.05;  
    ELSE  
        SET @v_discount = 0;  
  
    RETURN @v_discount;  
END;  
GO
```

```
220 SELECT dbo.fn_membership_discount(100000, 'Gold') AS discount_amount;  
221 GO  
222
```

100 % No issues found

Results Messages

	discount_amount
1	10000.00

```

/* =====
FUNCTION 4: Final Order Total
===== */

```

```
CREATE FUNCTION fn_order_final_total
```

```
(
```

```
    @p_order_id INT
```

```
)
```

```
RETURNS DECIMAL(10,2)
```

```
AS
```

```
BEGIN
```

```
    DECLARE @v_subtotal DECIMAL(10,2);
```

```
    DECLARE @v_tax DECIMAL(10,2);
```

```
    DECLARE @v_discount DECIMAL(10,2);
```

```
    DECLARE @v_membership VARCHAR(30);
```

```
    DECLARE @v_final_total DECIMAL(10,2);
```

```
    SELECT @v_membership = c.membership_level
```

```
    FROM orders o
```

```
    JOIN customers c ON o.customer_id = c.customer_id
```

```
    WHERE o.order_id = @p_order_id;
```

```
    SET @v_subtotal = dbo.fn_order_subtotal(@p_order_id);
```

```
    SET @v_tax = dbo.fn_tax_amount(@v_subtotal);
```

```
    SET @v_discount = dbo.fn_membership_discount(@v_subtotal, @v_membership);
```

```
    SET @v_final_total = @v_subtotal + @v_tax - @v_discount;
```

```
    RETURN ISNULL(@v_final_total, 0);
```

```
END;
```

```
GO
```

```

255     SELECT dbo.fn_order_final_total(1) AS final_order_total;
256     GO

```

100 %	✓ No issues found
Results	Messages
final_order_total	
1	1187500.00

```

/* =====
FUNCTION 5: Stock Status
===== */

```

```

CREATE FUNCTION fn_stock_status
(
    @p_stock_qty INT
)
RETURNS VARCHAR(30)
AS
BEGIN
    RETURN
    CASE
        WHEN @p_stock_qty = 0 THEN 'Out of Stock'
        WHEN @p_stock_qty < 10 THEN 'Low Stock'
        ELSE 'Available'
    END;
END;
GO

```

```

278 SELECT
279     product_name,
280     stock_qty,
281     dbo.fn_stock_status(stock_qty) AS stock_status
282 FROM products;
283 GO

```

100 %  No issues found

 Results  Messages

	product_name	stock_qty	stock_status
1	Laptop	10	Available
2	Mouse	100	Available
3	Keyboard	50	Available
4	Office Chair	20	Available
5	Desk	15	Available
6	Monitor	25	Available
7	Printer	8	Low Stock
8	USB Keyboard	120	Available
9	Gaming Mouse	75	Available
10	Bookshelf	12	Available
11	Meeting Table	5	Low Stock
12	Headphone	40	Available
13	Webcam	18	Available
14	Air Conditioner	3	Low Stock
15	Projector	6	Low Stock

```

/* =====
FUNCTION 6: Customer Order Count
===== */

```

```

CREATE FUNCTION fn_customer_order_count
(
    @p_customer_id INT
)
RETURNS INT
AS
BEGIN
    DECLARE @v_total_orders INT;

    SELECT @v_total_orders = COUNT(*)
    FROM orders
    WHERE customer_id = @p_customer_id;

    RETURN @v_total_orders;
END;
GO

```

```

306  SELECT
307      customer_name,
308      dbo.fn_customer_order_count(customer_id) AS total_orders
309  FROM customers;
310  GO

```

100 %  No issues found

 Results  Messages

	customer_name	total_orders
1	Aung Aung	1
2	Su Su	1
3	Kyaw Kyaw	1
4	Mya Mya	1
5	Ko Ko	1
6	Hla Hla	1
7	Zaw Zaw	1
8	Ei Ei	1
9	Tun Tun	1
10	May May	1

```

/* =====
FUNCTION 7: Membership Label
===== */

```

```

CREATE FUNCTION fn_membership_label
(
    @p_membership_level VARCHAR(30)
)
RETURNS VARCHAR(100)
AS
BEGIN
    RETURN
    CASE
        WHEN @p_membership_level = 'Gold' THEN 'Gold Member - 10% Discount'
        WHEN @p_membership_level = 'Silver' THEN 'Silver Member - 5% Discount'
        ELSE 'Normal Member - No Discount'
    END;
END;
GO

```

```

332  SELECT
333      customer_name,
334      membership_level,
335      dbo.fn_membership_label(membership_level) AS membership_description
336  FROM customers;
337  GO

```

100 % No issues found

Results Messages

	customer_name	membership_level	membership_description
1	Aung Aung	Gold	Gold Member - 10% Discount
2	Su Su	Silver	Silver Member - 5% Discount
3	Kyaw Kyaw	Normal	Normal Member - No Discount
4	Mya Mya	Gold	Gold Member - 10% Discount
5	Ko Ko	Silver	Silver Member - 5% Discount
6	Hla Hla	Normal	Normal Member - No Discount
7	Zaw Zaw	Gold	Gold Member - 10% Discount
8	Ei Ei	Silver	Silver Member - 5% Discount
9	Tun Tun	Normal	Normal Member - No Discount
10	May May	Gold	Gold Member - 10% Discount

```
/* =====  
FUNCTION 8: Shipping Fee  
===== */
```

```
CREATE FUNCTION fn_shipping_fee  
(  
    @p_amount DECIMAL(10,2)  
)  
RETURNS DECIMAL(10,2)  
AS  
BEGIN  
    IF @p_amount >= 500000  
        RETURN 0;  
  
    RETURN 5000;  
END;  
GO
```

```
357 SELECT dbo.fn_shipping_fee(300000) AS shipping_fee;  
358 GO
```

100 % No issues found

Results Messages

	shipping_fee
1	5000.00

```
/* =====  
FUNCTION 9: Grand Total  
===== */
```

```
CREATE FUNCTION fn_order_grand_total  
(  
    @p_order_id INT  
)  
RETURNS DECIMAL(10,2)  
AS  
BEGIN  
    DECLARE @v_final_total DECIMAL(10,2);  
    DECLARE @v_shipping_fee DECIMAL(10,2);
```

```

SET @v_final_total = dbo.fn_order_final_total(@p_order_id);
SET @v_shipping_fee = dbo.fn_shipping_fee(@v_final_total);

RETURN @v_final_total + @v_shipping_fee;

END;
GO

```

```

381 SELECT dbo.fn_order_grand_total(1) AS grand_total;
382 GO

```

100 % No issues found

Results Messages

	grand_total
1	1187500.00

```

/* =====
PRACTICAL REPORT QUERY
===== */

```

```

388 SELECT
389     o.order_id,
390     c.customer_name,
391     c.membership_level,
392     dbo.fn_order_subtotal(o.order_id) AS subtotal,
393     dbo.fn_tax_amount(dbo.fn_order_subtotal(o.order_id)) AS tax_amount,
394     dbo.fn_membership_discount(
395         dbo.fn_order_subtotal(o.order_id),
396         c.membership_level
397     ) AS discount_amount,
398     dbo.fn_order_final_total(o.order_id) AS final_total,
399     dbo.fn_shipping_fee(dbo.fn_order_final_total(o.order_id)) AS shipping_fee,
400     dbo.fn_order_grand_total(o.order_id) AS grand_total,
401     o.order_status
402 FROM orders o
403 JOIN customers c ON o.customer_id = c.customer_id;
404 GO

```

100 % No issues found

Results Messages

	order_id	customer_name	membership_level	subtotal	tax_amount	discount_amount	final_total	shipping_fee	grand_total	order_status
1	1	Aung Aung	Gold	1250000.00	62500.00	125000.00	1187500.00	0.00	1187500.00	Completed
2	2	Su Su	Silver	225000.00	11250.00	11250.00	225000.00	5000.00	230000.00	Completed
3	3	Kyaw Kyaw	Normal	250000.00	12500.00	0.00	262500.00	5000.00	267500.00	Pending
4	4	Mya Mya	Gold	1250000.00	62500.00	125000.00	1187500.00	0.00	1187500.00	Completed
5	5	Ko Ko	Silver	140000.00	7000.00	7000.00	140000.00	5000.00	145000.00	Pending
6	6	Hla Hla	Normal	200000.00	10000.00	0.00	210000.00	5000.00	215000.00	Cancelled
7	7	Zaw Zaw	Gold	905000.00	45250.00	90500.00	859750.00	0.00	859750.00	Completed
8	8	Ei Ei	Silver	190000.00	9500.00	9500.00	190000.00	5000.00	195000.00	Completed
9	9	Tun Tun	Normal	1800000.00	90000.00	0.00	1890000.00	0.00	1890000.00	Pending
10	10	May May	Gold	950000.00	47500.00	95000.00	902500.00	0.00	902500.00	Completed

```

/* =====
SHOW ALL USER FUNCTIONS
===== */

```

```

410  SELECT
411      name AS function_name,
412      create_date,
413      modify_date
414  FROM sys.objects
415  WHERE type IN ('FN', 'IF', 'TF')
416  ORDER BY name;
417  GO

```

100 % No issues found

Results

Messages

	function_name	create_date	modify_date
1	fn_customer_order_count	2026-05-20 19:47:51.687	2026-05-20 19:47:51.687
2	fn_membership_discount	2026-05-20 19:44:09.163	2026-05-20 19:44:09.163
3	fn_membership_label	2026-05-20 19:48:25.370	2026-05-20 19:48:25.370
4	fn_order_final_total	2026-05-20 19:45:09.563	2026-05-20 19:45:09.563
5	fn_order_grand_total	2026-05-20 19:49:43.183	2026-05-20 19:49:43.183
6	fn_order_subtotal	2026-05-20 19:41:55.163	2026-05-20 19:41:55.163
7	fn_shipping_fee	2026-05-20 19:49:11.053	2026-05-20 19:49:11.053
8	fn_stock_status	2026-05-20 19:46:57.007	2026-05-20 19:46:57.007
9	fn_tax_amount	2026-05-20 19:43:06.940	2026-05-20 19:43:06.940